

Anytime-Valid Screening of Agent Harness Components

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ABSTRACT

An LLM agent is a model inside a harness: a context policy, a tool set, a retry rule, a guardrail. Harness components move task success as much as model upgrades do, and the affordable ways to decide which components stay fail on measured parameters: a Resolution III fractional design, the textbook cheap screen, attributes a two-factor interaction to an inert component in 94.2% of replications; unadjusted interim looks inflate family-wise error to 41.8% at ten components under the global null; and the point-comparison acceptance used by self-improvement loops false-adopts with probability 0.999 per round, computed exactly. We propose screening with independently randomised levels and one e-process per component. A single supermartingale argument proves family-wise error control at any stopping time under the intersection null over reachable committed contexts; the worst measured admission is 0.2% across the 18 cells built with that null true. Pairing each run with its exact mirror on the same task is a variance option, not an identification device; an exact per-invocation log-growth criterion decides when the pairing pays, and matches every resolved cell of our simulation grid. A live screen of five components of a production agent runtime pruned one guardrail under this guarantee, and an independent fixed-sample comparison of the two committed configurations, on fresh draws after the decision, checked it. An open-source implementation and every artefact accompany the paper.

Manuscript, artefacts and figures. [PDF](#) · [exp_screen.json](#) · [exp_mirror.json](#) · [targets.json](#) · [citations.json](#) · [tables](#). Every number below is read from these files at build time.

Two ways a cheap screen names a component that does nothing

left: an inert component admitted when two real components interact, $k=5$. right: family-wise error under the global null, $k=10$. 6400 invocations, 400 replications.

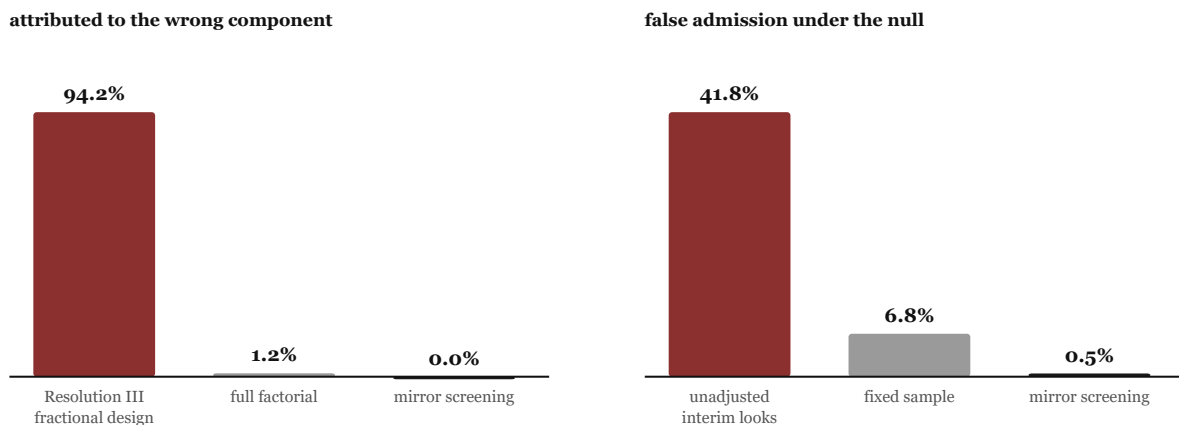


Figure 1. The two failures this paper removes, measured on a simulator calibrated to a production deployment. Left: with two components genuinely interacting (synergy scenario, heterogeneous tasks, 6400 invocations), a Resolution III fractional design admits the inert component that carries their alias in 94.2% of replications. Right: under the global null at ten components, unadjusted interim looks raise the family-wise error rate to 41.8%. The randomised screen stays below the nominal level in both settings at the same budget.

1. The screening problem

An agent that resolves a software issue (Jimenez et al., 2023) or drives a terminal is not a model. It is a model inside a harness that decides what the model sees, which tools it may call, when a failed step is retried, and when the episode ends. The size of that layer is no longer in doubt: a recent runtime raises a fixed model from 83.1 to 92.1 on Terminal-Bench 2.1 by changing the execution system alone, and to the 95.3 of its title with its largest harness configuration (Qin et al., 2026); a dedicated benchmark now measures harness effects across models (Yao et al., 2026). Once a team accepts that the harness matters, it acquires a portfolio of switches, and with it the question this paper is about: which of them are actually paying, and which are inherited superstition.

The question is a screening problem, and the field currently answers it in one of two ways. The careful way is to enumerate the full factorial and average: the closest prior study runs the complete 2^5 over five scaffolding components with up to ten seeds per cell, and reports that stacking components can hurt (Liu, 2026). The common way is to run a partial sweep and stop when the numbers look decided, a temptation that grows with the price of a rollout; a replay study of public benchmarks finds that the fraction of tasks needed before a partial run supports the completed decision ranges from 15% to more than 90% depending on the benchmark (Huang, 2026). Neither way carries an error guarantee for the decision it is used to make.

Two properties of agent evaluation make this worse than it looks. First, outcomes are not stable: replaying identical inputs through the deployment used in this paper flips 11.7% of row-level verdicts, and consistency under semantically preserving perturbation is now itself an object of study (Raj et al., 2026). Second, the components interact, so an estimator that confounds a main effect with an interaction will not merely be imprecise, it will name the wrong component. We show below that this matters for the design a cost-pressed team would reach for: a Resolution III fractional design, the textbook answer to enumeration cost in industrial screening (Box, Hunter and Hunter, 2005) though not yet documented in harness studies, attributes an interaction to an inert component in nine cases out of ten.

This paper proposes a procedure with three properties that the current practice does not have together. It does not enumerate configurations, so its cost is independent of the number of components. It cannot confound a two-factor interaction with a main effect, by construction rather than by design resolution. And it is valid under continuous monitoring, so a team may look at the evidence whenever it likes and stop whenever it likes, which is what teams do. The price is quantified rather than hidden: a few points of power at equal budget against a fixed-sample full factorial, decomposed exactly in Section 6.

Concretely, the paper contributes: (i) a three-line screening procedure whose cost per row is two invocations regardless of the number of components (Section 3); (ii) a supermartingale proof of family-wise error control at any stopping time under the intersection null over reachable committed contexts, and a composition result showing a self-improvement loop inherits the judge's guarantee whatever its proposer does (Propositions 4 and 5); (iii) an exact per-invocation log-growth criterion that predicts, cell by cell, when mirror pairing beats unpaired screening (Section 6); and (iv) a measured account of how the two standard alternatives fail, together with a live deployment run in which the procedure pruned one guardrail at its harm process's α -level, with an independent post-decision confirmation reported alongside (Section 7).

2. Related work and what is left

Harness effects. Yao et al., 2026 measures them across models; Chen et al., 2026 isolates scaffold components through controlled composition; Liu, 2026 reports destructive interaction between components under a full factorial; Qin et al., 2026 demonstrates how large the effect of a runtime can be. All four establish that component attribution is worth doing. None supplies a decision rule with an error guarantee, which is what a team needs before it removes a component from a production harness.

Evaluation methodology. Kapoor et al., 2024 makes cost a first-class axis and Kapoor et al., 2025 supplies the infrastructure to compare agents under a cost budget. Miller, 2024 gives fixed-sample confidence intervals for evaluations, and Zhu et al., 2025 catalogues the ways agentic benchmarks are built wrongly. Our procedure is complementary: it inherits the reporting standards of that literature and adds a stopping rule that survives the analyst looking early.

What we do not claim as new. The mirror difference is an old device. Antithetic variates pair a draw with its reflection to reduce variance (Hammersley and Morton, 1956); simultaneous perturbation stochastic approximation perturbs every coordinate at once with Rademacher signs and takes a two-sided difference (Spall, 1992); mirrored sampling does the same inside evolution strategies; and folding a Resolution III design to reach Resolution IV is textbook (Box, Hunter and Hunter, 2005). Our contribution is not the estimator. It is the combination of that estimator with a per-component e-process, the resulting family-wise guarantee under optional stopping, the application to harness screening, and the measurement of how badly the standard alternatives fail on this problem.

Anytime-valid inference. The sequential tools are likewise not ours: power-one tests (Robbins, 1970), testing by betting (Shafer, 2019), safe tests (Grünwald et al., 2019), time-uniform confidence sequences (Howard et al., 2018), betting-based mean estimation whose plug-in bet family we adopt (Waudby-Smith et al., 2020), always-valid inference for A/B peeking, the industrial ancestor of our second headline failure (Johari et al., 2015), and the survey that fixes our terminology (Ramdas et al., 2022). The false-discovery-rate extension named in Section 8 is e-BH (Wang et al., 2020). Reading main

effects off randomly drawn factor levels also has prior art and a published critique: random balance experimentation (Satterthwaite, 1959), whose defect was precisely the absence of an error-controlled decision rule on top of the random design; the e-process is that missing rule. Best-arm identification (Jamieson et al., 2013; Kaufmann et al., 2014) answers a different question, selecting the best of k alternatives rather than testing each of k components against its own null, and does not handle the paired factorial structure here. What none of this supplies is the application: a screening rule in which the randomisation, the pairing and the intersection null are matched to how agent teams commit component decisions mid-run.

Table 1. Prior work cited in this section, resolved against the arXiv API at build time. A stale or wrong identifier fails the build.

reference	title	first posted
Jamieson et al., 2013	lil' UCB : An Optimal Exploration Algorithm for Multi-Armed Bandits	2013-12-27
Kaufmann et al., 2014	On the Complexity of Best Arm Identification in Multi-Armed Bandit Models	2014-07-16
Johari et al., 2015	Always Valid Inference: Bringing Sequential Analysis to A/B Testing	2015-12-15
Howard et al., 2018	Time-uniform, nonparametric, nonasymptotic confidence sequences	2018-10-18
Shafer, 2019	The Language of Betting as a Strategy for Statistical and Scientific Communication	2019-03-16
Grünwald et al., 2019	Safe Testing	2019-06-18
Wang et al., 2020	False discovery rate control with e-values	2020-09-06
Waudby-Smith et al., 2020	Estimating means of bounded random variables by betting	2020-10-19
Ramdas et al., 2022	Game-theoretic statistics and safe anytime-valid inference	2022-10-04
Jimenez et al., 2023	SWE-bench: Can Language Models Resolve Real-World GitHub Issues?	2023-10-10
Kapoor et al., 2024	AI Agents That Matter	2024-07-01
Miller, 2024	Adding Error Bars to Evals: A Statistical Approach to Language Model Evaluations	2024-11-01
Zhu et al., 2025	Establishing Best Practices for Building Rigorous Agentic Benchmarks	2025-07-03
Agrawal et al., 2025	GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning	2025-07-25
Kapoor et al., 2025	Holistic Agent Leaderboard: The Missing Infrastructure for AI Agent Evaluation	2025-10-13
Merrill et al., 2026	Terminal-Bench: Benchmarking Agents on Hard, Realistic Tasks in Command Line Interfaces	2026-01-17
Liu, 2026	More Is Not Always Better: Cross-Component Interference in LLM Agent Scaffolding	2026-05-07
Raj et al., 2026	Consistency as a Testable Property: Statistical Methods to Evaluate AI Agent Reliability	2026-05-11
Yao et al., 2026	Harness-Bench: Measuring Harness Effects across Models in Realistic Agent Workflows	2026-05-27
Chen et al., 2026	AgentSpec: Understanding Embodied Agent Scaffolds Through Controlled Composition	2026-06-12
Huang, 2026	How Many Tasks Are Enough for Agent Benchmark Decisions? A Replay Analysis of Public LLM Agent ...	2026-07-14
Qin et al., 2026	StateM: Reaching 95.3% Raw Accuracy, or a \$15 Frontier Run, on Terminal-Bench 2.1 via Harness S...	2026-08-15

3. Mirror screening

Write a harness configuration as a vector x in $\{-1, +1\}^k$, one coordinate per component, where $+1$ means the component is switched on. Let $y(x, t)$ be the verified outcome of running the agent on task t under configuration x , and let the response decompose as a polynomial in x with arbitrary coefficients:

$$y(x, t) = \mu(t) + \sum_i \beta_i x_i + \sum_{i < j} \gamma_{ij} x_i x_j + \sum_{i < j < l} \tau_{ijl} x_i x_j x_l + \dots + \varepsilon$$

The procedure is three lines. For each row: draw a task t and draw the levels x independently and uniformly from $\{-1, +1\}$; execute the agent twice on the same task, once at x and once at the mirror $-x$; form the paired difference $d = (y(x, t) - y(-x, t)) / 2$ and the per-component signal $g_i = x_i d$. Components already decided are held at their committed level in both

executions, so they contribute nothing to the difference. Each component carries two test supermartingales, one betting that the component helps and one on the negation, updated with a bet size computed only from earlier rows:

ALGORITHM 1. MIRROR SCREENING

1. Input: components $1..k$, task pool T , level α , invocation budget B .
2. State: for each component, two e-values initialised at one, and a decision flag.
3. Repeat until the budget is spent or every component is decided:
4. draw a task t from T and levels x uniformly from $\{-1, +1\}^k$; overwrite the coordinates of decided components with their committed level,
5. run the agent on t at x and at the mirror of x , where the mirror flips only the undecided coordinates,
6. form $d = (y(x, t) - y(\text{mirror}, t)) / 2$ and $g_i = x_i d$,
7. update both e-values of every undecided component with a predictable non-negative clipped bet; admit when the for-process crosses k/α , prune when the against-process crosses k/α , retire as undecided-inactive (not pruned, no error statement) when the for-process falls below the futility floor $f = 0.05$.
8. Output: the admitted set, the pruned set, the undecided-inactive set, and the evidence path of each component.

Nothing else is required. There is no design matrix to build, no resolution to check, no sample size to commit to in advance, and no enumeration of the 2^k configurations. The cost is two model invocations per row regardless of k , and the analyst may inspect E_i at any moment and stop at any moment.

An earlier construction paired each run with a fixed all-off champion instead of the mirror: unbiased, and it too cancels the task effect, but half of its budget re-runs an uninformative baseline and its signal carries the other components' effects at full variance where the mirror halves them (neither cancels them; those terms are odd). Section 6 prices this: the champion-paired arm recovers a fraction of what mirror pairing recovers at equal budget.

4. Why the procedure is safe, and exactly what is guaranteed

PROPOSITION 1 (ANNIHILATION OF EVEN-ORDER TERMS). For any coefficients and any x , the mirror difference $d(x) = (y(x) - y(-x)) / 2$ equals $\sum_i \beta_i x_i + \sum_{i < j < l} \tau_{ijl} x_i x_j x_l + \dots$: every even-order term is removed exactly, including the intercept, the task effect, and all two-factor interactions.

Proof. Substituting $-x$ into a monomial of order m multiplies it by $(-1)^m$; even monomials cancel in the difference, odd ones survive. The task effect is even of order zero, which is why the two executions must share a task. $\square$

PROPOSITION 2 (UNBIASED SIGNAL UNDER FULL MIRRORING). While no component has been decided, the levels are iid Rademacher and every coordinate is mirrored, so $E[g_i] = \beta_i$ exactly: each surviving cross term contains at least two distinct independent signs and has mean zero.

PROPOSITION 3 (WHAT THE E-PROCESS GUARANTEES, STATED FOR THE ALGORITHM ACTUALLY RUN). The betting strategy is the plug-in rule $\lambda = \text{clip}(\hat{\mu} / (\hat{\sigma}^2 + \hat{\mu}^2 + c), 0, 0.45)$ with running estimates from earlier rows only, a stabilised empirical-mean bet in the GRAPA family of Waudby-Smith et al., 2020, with stabiliser $c = 0.1$ and $\lambda = 0.1$ for the first two rows before running estimates exist. It is computed from earlier rows only, and is clipped to $[0, 0.45]$ while $|g_i| \leq 1$, so $1 + \lambda_i g_i > 0$ and each E_i is a non-negative process with $E[E_i(n) \mid \text{past}] \leq E_i(n-1)$ under its null. Ville's inequality then bounds the chance that E_i ever reaches k/α by α/k , and a union bound over components gives family-wise error at most α at any stopping time. Two separate families run at once: the admission processes control admitting a component whose effect is not positive, and the harm processes control pruning a component whose effect is not negative, each at level α ; a reader who acts on both admissions and prunings is carrying 2α in total, and every table reports the two families separately against their own α . The futility floor ($E_i \leq 0.05$) is a budget device, not part of either guarantee: a component stopped for futility is reported as undecided-inactive, and no error statement is attached to that outcome.

PROPOSITION 4 (THE INTERSECTION NULL, WHICH IS THE THEOREM THIS PAPER IS ABOUT). For component i and row n let $\theta_i^{(n)}$ denote the conditional mean of its paired increment given the past, that is, the main effect of i conditional on the levels committed so far. Define the intersection null $H_0^\cap = \cap_n \{\theta_i^{(n)} \leq 0\}$: the component helps in no reachable committed context. Under $H_0^\cap$, with the predictable non-negative clipped bet of Proposition 3, every factor $1 + \lambda_i g_i$ has conditional mean at most one at every row, in every epoch, including across commitment times, so E_i is a non-negative supermartingale on the whole run and Ville's inequality bounds the probability of ever admitting i by α/k ; a union bound over components gives family-wise error at most α at any stopping time. No restart, no epoch bookkeeping and no correction for the adaptivity of the commitments is needed, because the null itself quantifies over every reachable context.

Proof. The commitment times are stopping times of the same filtration, the committed levels s_c are measurable at commitment, and the levels of the undecided coordinates of row n are drawn independently of the past, so $E[g_i \mid \text{past}] = \theta_i^{(n)} \leq 0$ under the null at every n . With $\lambda_i \geq 0$ predictable and $\lambda_i |g_i| < 1$, $E[E_i(n) \mid \text{past}] = E_i(n-1)(1 + \lambda_i \theta_i^{(n)}) \leq E_i(n-1)$. Ville and the union bound do the rest. $\square$

Two readings follow, and both are measured in Section 6. A component outside $H_0^\cap$ because its effect changes sign, positive before a commitment and negative after, carries no guarantee: its admission rate is the price of the flip, not an error rate of the test. And a component inside $H_0^\cap$, non-positive at every stage, must be admitted at most α often, which is the rate the theorem actually promises. A deployer who instead wants a guarantee stated at the final committed configuration might restart the undecided processes at each commitment and spend α across epochs; both variants are run, and the measurement in Section 6 shows the restart is not the repair it looks like.

Two attributions must be kept apart, because the natural reading of the construction gets them wrong. Removing alias bias does not require the mirror at all: drawing the levels independently already makes every interaction term mean-zero in any main-effect contrast, which is why the randomised unpaired baselines in Section 6 show no alias inflation. What randomisation does not remove is the task effect $\mu(t)$, which stays inside every unpaired increment and inflates its variance. The mirror removes it row by row. The pairing is therefore a variance device, not a bias device, and its value must grow with the variance of task difficulty and vanish without it; Section 6 reports exactly that sweep.

PROPOSITION 5 (THE COMPOSED LOOP INHERITS THE JUDGE'S GUARANTEE). Consider a loop in which a proposer emits candidate components, an executor produces the paired outcomes, and this procedure judges. Suppose each candidate's identity is fixed before the first row of its evidence is observed, and the executor's paired outcomes for a row are exchangeable given the task. Then, however adaptively the proposer chose the candidates, and however the analyst monitors, the probability that any candidate with non-positive conditional effect is ever admitted in the round is at most α . The proposer's adaptivity cannot leak into the error rate because each e-process starts at one after its candidate is fixed and its bets are predictable; the executor cannot leak bias because the mirror difference removes everything the pair shares. The loop's acceptance error is the judge's alone.

The counterpart is computed exactly rather than simulated. Under the calibrated null a point comparison adopts a do-nothing candidate with probability $q = (1 - P(\text{tie}))/2$, which at 320 rows per side equals 0.484; with ten candidates the round-level failure is $1 - (1 - q)^{10} = 0.999$, and with five, 0.965 at the corresponding budget. The simulated null rates in Table 9 agree within sampling error: $96.8 \pm 0.9\%$ at five candidates against the exact 96.5%, and $100.0 (\geq 99.3)\%$ at ten against the exact 99.9%. A loop that accepts on point comparisons is broken by arithmetic, not by bad luck.

The contrast with subsetting a design remains as before: retaining only the rows of a Resolution III design on which a decided component is off correlates the surviving columns and leaks a live effect into an inert one (0.4000 to 0.1000 in the enumeration), while the estimator that keeps the full matrix and only forces the executed configuration does not.

5. Experimental setup

The evaluation isolates one axis at a time, because a comparison that changes the design, the pairing and the inference rule at once cannot attribute its own result. Every arm receives an identical number of model invocations. The arms are: the full factorial with a fixed-sample z test and a one-sided Bonferroni threshold over k , which is current practice; a Resolution III

fractional design with the same test; the same design with unadjusted interim analysis every forty rows, stopping at the first crossing, which is common practice; the champion-paired e-process; and mirror screening, with and without the sequential rule.

Rows are drawn from a fixed pool of 200 tasks, the order of magnitude of Terminal-Bench 2.0, which contains 89 verified terminal tasks (Merrill et al., 2026). Each task carries a latent difficulty drawn from a Beta law, so that assigning tasks to configurations without pairing incurs the composition noise that a real benchmark suite incurs. Switching on an active component removes a share of the remaining failure mass; when two active components are on together the pass probability shifts by a fixed interaction, run at both signs because stacking has been reported to hurt as well as help (Liu, 2026). Every outcome is finally mixed toward a coin flip at the replay flip rate measured on our own deployment.

Table 2. Parameters. The first two are measurements from a production deployment, not choices; the rest are swept.

parameter	value	source
replay flip rate	0.117	same-input replay on the deployment
champion failure share	0.509	same deployment
task pool	200	order of Terminal-Bench 2.0 (89 tasks)
components k	5 and 10	5 matches the prior full-factorial study; at 10 the full factorial needs 1024 configurations
active components	2	the remainder are inert by construction
interaction	-0.12 and +0.12	both signs
budget	400, 1600, 6400 invocations	2x, 8x and 32x the task pool
alpha	0.05, Bonferroni over k	convention
replications	400 per cell	standard errors below 2 points

Table 2 lists every parameter with its provenance. Power is meaningless without the size of the effect being detected, so the estimand is computed exactly rather than described. Enumerating all configurations gives a true mirror contrast of 0.0312 for an active component under main effects alone, 0.0577 when the positive interaction is present, and 0.0047 when the negative interaction is present, which is close enough to zero that no method should detect it at these budgets. The true contrast of every inert component is exactly zero in all scenarios, to the last digit the enumeration carries.

6. Results

A fractional design names the wrong component

When two active components interact, the Resolution III design admits the inert component that carries their alias in 94.2% of replications at $k = 5$ and 88.2% at $k = 10$, at the same budget where the full factorial, the fixed-sample fold-over and every randomised arm stay near the nominal level. The recovered contrasts in Figure 2 show why: the aliased component is credited with the entire interaction, and no amount of extra sampling removes it, because the bias is in the estimator rather than in its variance.

Table 3. Interaction present, 6400 invocations, 400 replications. The last column counts replications in which the component that carries the alias of the two active components was admitted.

method	recovery, $k=5$	FWER, $k=5$	alias admitted, $k=5$	alias admitted, $k=10$
full factorial, fixed sample	100.0 (≥ 99.3)	3.2 ± 0.9	$1.2 \pm 0.5\%$	$0.8 \pm 0.4\%$
Resolution III, fixed sample	100.0 (≥ 99.3)	94.8 ± 1.1	$94.2 \pm 1.2\%$	$88.2 \pm 1.6\%$
fold-over (Resolution IV), fixed sample	100.0 (≥ 99.3)	2.8 ± 0.8	$1.0 \pm 0.5\%$	$0.2 (\leq 1.2)\%$
fold-over, unadjusted interim looks	47.2 ± 0.9	10.5 ± 1.5	$3.5 \pm 0.9\%$	$2.2 \pm 0.7\%$
champion paired, e-process	91.2 ± 1.2	2.2 ± 0.7	$0.8 \pm 0.4\%$	$0.5 (\leq 1.6)\%$
randomised levels, unpaired, fixed sample	100.0 (≥ 99.3)	3.8 ± 0.9	$1.2 \pm 0.5\%$	$0.5 (\leq 1.6)\%$
randomised levels, unpaired, e-process	100.0 (≥ 99.3)	1.2 ± 0.6	$0.8 \pm 0.4\%$	$0.0 (\leq 0.7)\%$

mirror pairing, fixed sample	100.0 (≥ 99.3)	1.8 ± 0.7	$1.0 \pm 0.5\%$	$1.0 \pm 0.5\%$
mirror pairing, e-process (this paper)	100.0 (≥ 99.3)	$0.0 (\leq 0.7)$	$0.0 (\leq 0.7)\%$	$0.0 (\leq 0.7)\%$

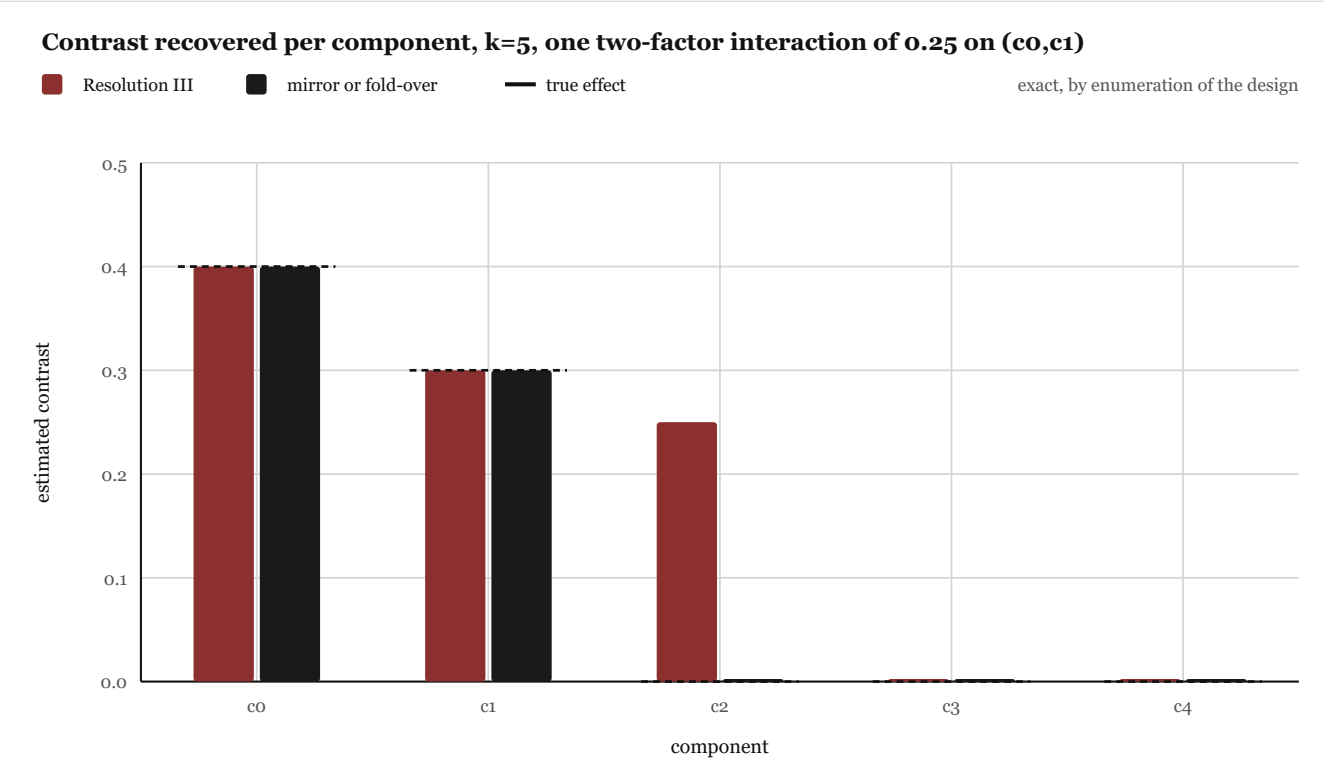


Figure 2. Contrast recovered per component with one two-factor interaction of 0.25 between components 0 and 1, computed exactly by enumerating each design. The Resolution III design credits the whole interaction to inert component 2; the mirror difference and the fold-over design return the true effect on every component.

Looking early breaks a fixed-sample threshold

Under the global null, testing every forty rows against a fixed-sample Bonferroni threshold and stopping at the first crossing raises the family-wise error rate to 29.0% at five components and 41.8% at ten, against a nominal 5%. The e-process arms stay at 0.0% and 0.5%. This is not a subtle effect and it does not require bad faith: the interim rule is what a team does when each rollout costs money and the numbers already look decided. Figure 3 shows the inflation by budget and component count, and Table 13 tabulates every cell behind it.

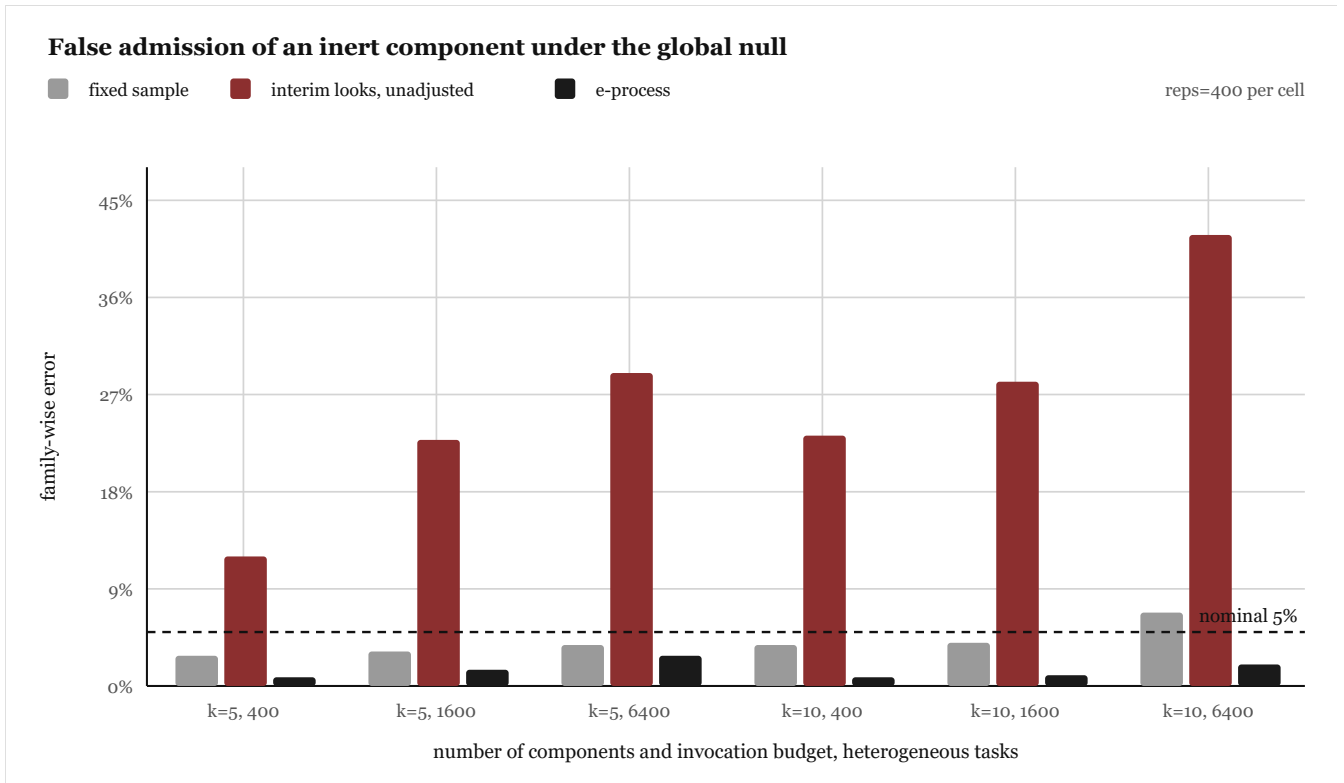


Figure 3. False admission of an inert component under the global null, by budget and number of components. Unadjusted interim analysis is far above the nominal level everywhere; the e-process is below it everywhere.

The price of validity, paid in samples and measured

At an equal budget the fixed-sample full factorial recovers 99.6% of the active components at $k = 5$ and the sequential mirror procedure recovers 91.8%; at $k = 10$ the figures are 99.1% and 89.6% (Figure 4; Figure 5 repeats the sweep at $k = 10$). That gap is what anytime validity costs on this problem. Against the construction this work replaces, the champion-paired e-process recovers 19.8% and 16.9% under the same conditions, power ratios of 4.6x and 5.3x explained by the half of its budget spent on an uninformative baseline; a new adopter should compare against the unpaired default of Table 5, which this arm is not.

The gap mixes two prices, and a threshold sweep separates them. The sweep is a separate experiment (exp_v2.json, 400 replications per point, its own seeds), so its nominal- α cell reads 92.4% where Table 4's independent grid reads 91.8 ± 1.0 - sampling error between grids, quoted so the reader need not reconcile them. Raising α until the e-process arm's realised null error matches the fixed-sample arms' ($\alpha = 0.35$, realised 4.8% against the full factorial's 4.5% in the same grid) raises its recovery to 97.9%: most of the gap is the slack of the union bound and Ville's inequality, recoverable by calibration if a team is willing to state its guarantee at the realised rather than the nominal level, and the remainder, about 1.7 points, is the price of the anytime property itself.

Table 14. The calibration sweep behind the decomposition (mirror e-process, $k = 5$, heterogeneous, 6400 invocations, 400 replications per row) with the fixed-sample full factorial of the same grid as comparator, plus the auxiliary cells quoted in the text. Three similar-looking numbers are three different cells: this table's comparator FWER is the mirror grid's global null; Table 4's 3.5% for the same arm is its inert-admission rate in the main-effect cell; Table 13's 4.8 ± 1.1 is the design-grid replication of the global null. Recovery is monotone in α ; the realised null error stays below nominal throughout, which is the union-bound and Ville slack the decomposition prices.

arm	alpha	recovery, main cell	realised FWER, null cell
mirror e-process	0.05	92.4	0.2
mirror e-process	0.10	94.6	2.0
mirror e-process	0.20	97.6	1.8
mirror e-process	0.35	97.9	4.8
mirror e-process	0.50	97.9	8.2
full factorial, fixed sample	0.05 (Bonferroni)	99.6 (≥ 98.4)	4.5 ± 1.0

mirror pairing, fixed sample (1600 invocations)	0.05 (Bonferroni)	62.6 ± 1.7	-
randomised unpaired, fixed sample (1600 invocations)	0.05 (Bonferroni)	58.5 ± 1.7	-
live pilot chooser, raw / corrected ratio	-	1.39 / 1.22	-

Table 4. Main effects only, 6400 invocations, 400 replications. Standard errors here and in every simulation table are computed at the replication level: the recovery fraction of a replication is one observation, never its components separately. For these cells the naive per-component binomial standard error would read 0.1-0.3 points smaller; the numbers shown are the clustered ones.

method	recovery, k=5	FWER, k=5	recovery, k=10	FWER, k=10
full factorial, fixed sample	99.6 (≥98.4)	3.5 ± 0.9	99.1 ± 0.3	4.0 ± 1.0
Resolution III, fixed sample	99.9 (≥99.3)	3.2 ± 0.9	99.0 ± 0.4	2.8 ± 0.8
fold-over (Resolution IV), fixed sample	99.8 (≥98.8)	3.0 ± 0.9	99.6 (≥98.4)	3.8 ± 0.9
fold-over, unadjusted interim looks	43.2 ± 0.9	16.5 ± 1.9	39.6 ± 1.1	22.8 ± 2.1
champion paired, e-process	19.8 ± 1.4	1.8 ± 0.7	16.9 ± 1.3	1.8 ± 0.7
randomised levels, unpaired, fixed sample	99.9 (≥99.3)	3.5 ± 0.9	99.6 (≥98.4)	4.8 ± 1.1
randomised levels, unpaired, e-process	92.6 ± 0.9	1.0 ± 0.5	89.8 ± 1.1	1.5 ± 0.6
mirror pairing, fixed sample	99.9 (≥99.3)	1.5 ± 0.6	99.9 (≥99.3)	4.5 ± 1.0
mirror pairing, e-process (this paper)	91.8 ± 1.0	0.2 (≤1.2)	89.6 ± 1.1	1.0 ± 0.5

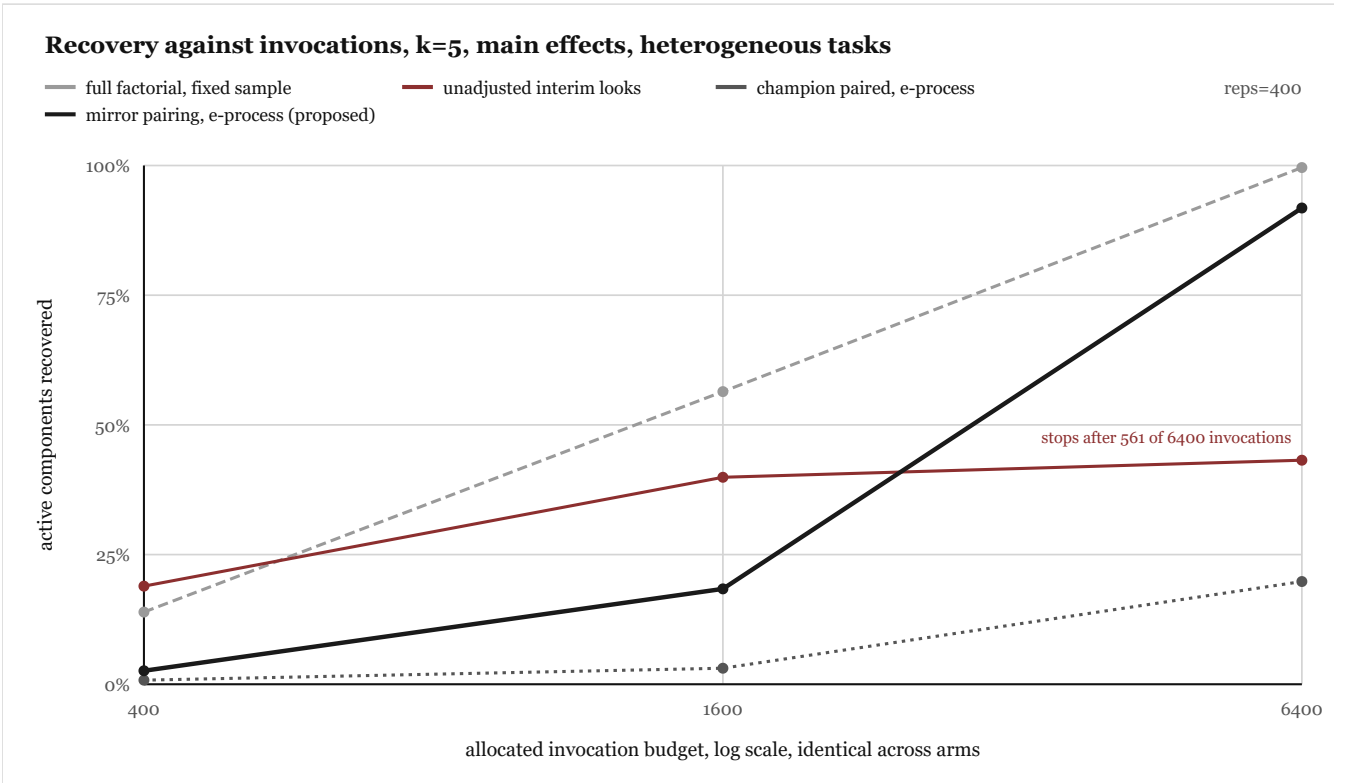


Figure 4. Recovery against invocations at k = 5. The interim arm consumes far fewer invocations because it stops early, which is exactly why its error rate is inflated.

Recovery against invocations, $k=10$, main effects, heterogeneous tasks

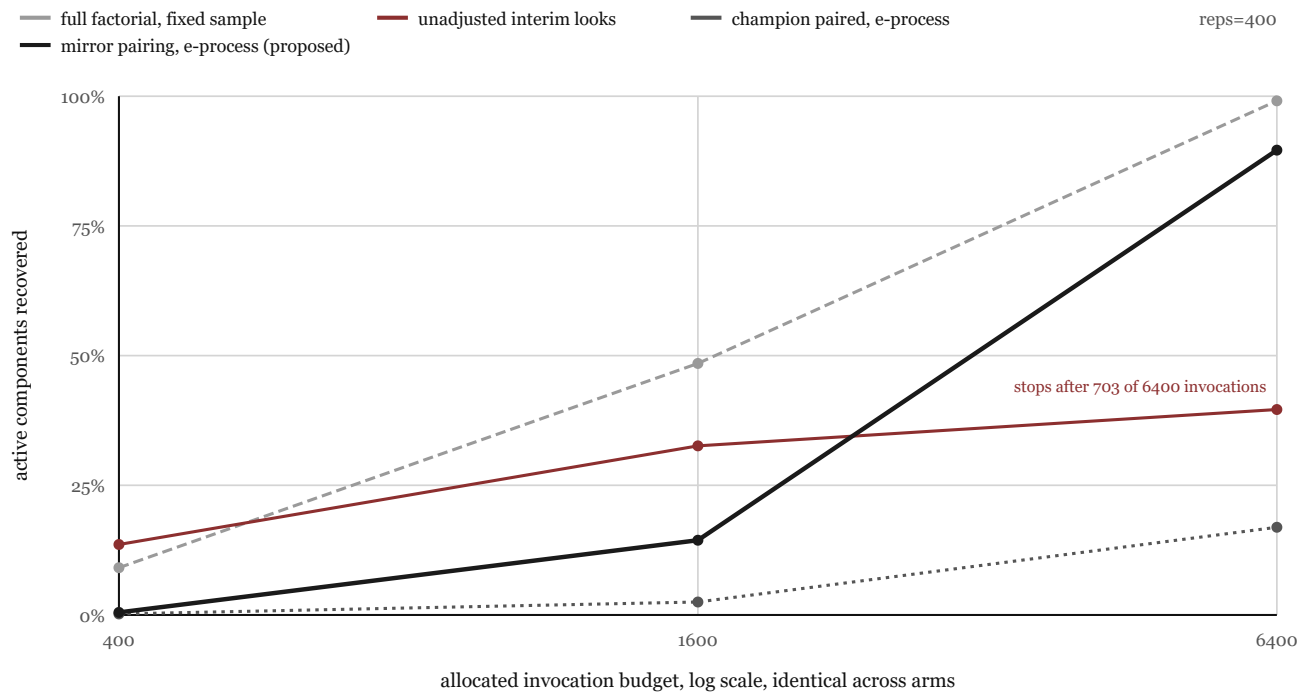


Figure 5. The same at $k = 10$, where enumerating the full factorial would require 1024 configurations. The cost of mirror screening does not grow with the number of configurations, only the threshold k/α does.

Evidence path per component in a single run

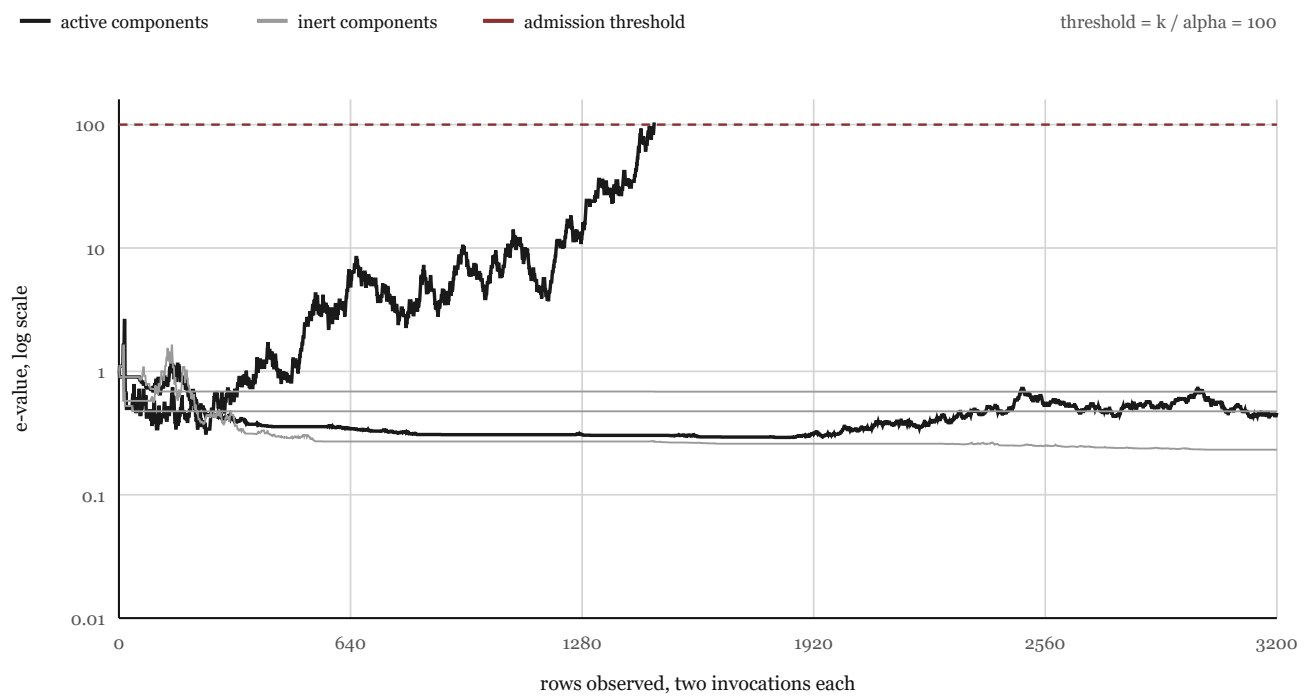


Figure 6. Evidence paths in a single run. One active component crosses the admission threshold and the run stops for that component; the second, with a smaller effect, has not accumulated enough evidence and remains undecided. Inert components drift downward.

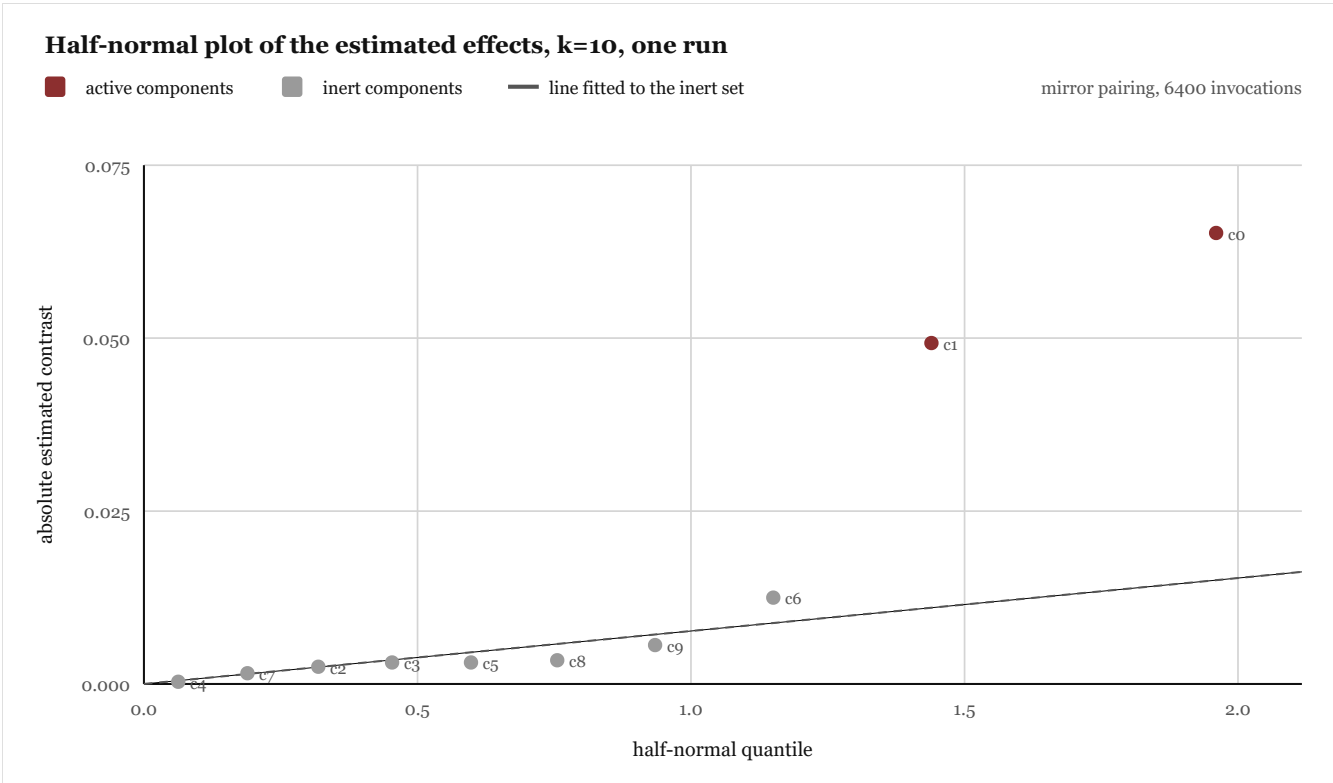


Figure 7. The standard screening plot for one run at ten components: absolute estimated contrasts against half-normal quantiles. The eight inert components lie on a line through the origin, the two active ones leave it. This is the diagnostic a practitioner reads before deciding what to keep.

What randomisation buys and what the mirror buys, separated

The reviewer-facing question is whether the mirror is doing anything that randomising the levels does not already do. For bias, the answer is no, and the randomised unpaired arms show it: their alias admission sits at the nominal level in every cell. For power, the answer depends on exactly one quantity, the spread of task difficulty, because the task effect is what the mirror removes from each increment. With homogeneous tasks the unpaired e-process matches the mirrored one at half the invocations per row; as the difficulty spread grows the unpaired increments carry that spread as noise and the mirrored procedure pulls away. Table 5 and Figure 8 report both at their own invocation accounting.

Table 5. Paired against unpaired at $k = 5$, main effects, 6400 invocations, by task-difficulty spread. The mirror pays when tasks differ; it is unnecessary when they do not. The enumerated estimand is the same in every row to three decimals (0.0312, 0.0312, 0.0311), because the spreads share the same mean failure mass: the movement across rows is variance, not estimand.

task difficulty	mirror e-process, recovery	unpaired e-process, recovery	mirror FWER	unpaired FWER
homogeneous	87.1 $\pm$ 1.1	93.8 $\pm$ 0.9	0.8 $\pm$ 0.4	1.2 $\pm$ 0.6
Beta, concentration 4	91.8 $\pm$ 1.0	92.6 $\pm$ 0.9	0.2 ($\leq$ 1.2)	1.0 $\pm$ 0.5
Beta, concentration 1	96.9 $\pm$ 0.6	92.5 $\pm$ 0.9	0.0 ($\leq$ 0.7)	1.5 $\pm$ 0.6

The mirror pays only when tasks differ: recovery by task-difficulty spread, $k=5$

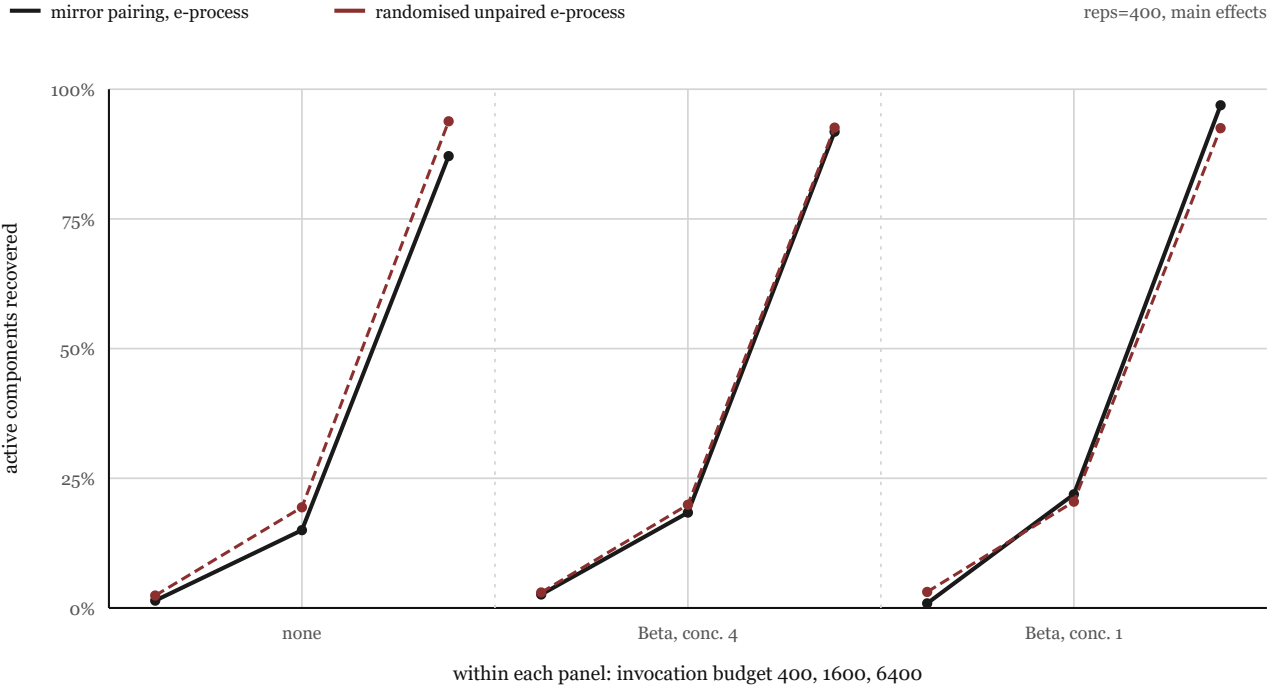


Figure 8. Recovery of the two active components by task-difficulty spread, $k = 5$, main effects. With homogeneous tasks the unpaired procedure matches the mirror at half the invocations per row; as the spread grows the task effect stays inside every unpaired increment and the mirror pulls away.

The sweep has a model behind it, and the naive version of that model is wrong in an instructive way. At a fixed sample size the ordering is settled by variance alone, and the below-ceiling budget confirms it: at 1600 invocations the mirror fixed-sample arm recovers 62.6% against 58.5% unpaired, the direction the per-sample ratio predicts. The e-process reverses that ordering on homogeneous tasks, and the reason is the bet, not the variance. Per sample, the mirrored increment has variance $p(1-p)/2$ against $E[p(1-p)] + \text{Var}_t(p)$ unpaired, a ratio of 1.00, 1.19 and 1.64 across the three spreads; but the mirror spends two invocations per update, and the stabilised bet $\lambda = \mu/(\sigma^2 + \mu^2 + c)$ rewards the lower-variance arm less than proportionally. Computing the expected log-growth of each e-process per invocation exactly, from the calibrated increment distributions with the actual clipped bet, gives growth ratios mirror to unpaired of 0.871, 0.964 and 1.136 across the three spreads: the mirror is predicted to lose on homogeneous tasks, to roughly tie at our calibrated heterogeneity, and to win only at strong spread. Two disclosures temper the agreement with Table 5. The criterion and the simulator share the generative model, so these checks are model-consistency, not external validation; and in the concentration-4 binary cell the measured difference (91.8 against 92.6, difference standard error near 1.4 points) is within noise, so the criterion's sign is resolved in six of the seven paired-versus-unpaired cells and consistent with a tie in the seventh. The practitioner's rule is to estimate the between-task spread of the champion's pass rate from a pilot, evaluate this ratio, and choose the arm; on graded verifiers the Bernoulli floor vanishes and the same computation makes the mirror decisive, which Table 6 confirms.

Table 6. Graded verifier (score in $[0,1]$, measurement noise $\sigma = 0.05$), $k = 5$, budgets as marked per row, 400 replications, standard errors at the replication level. The estimand is the enumerated contrast of the binary tables, 0.0312 per active component. The homogeneous row sits at 400 invocations, where the growth model's negative prediction for the mirror is genuinely at risk and holds; the same model, computed on the graded increments, puts the invocations-to-threshold at 840 for the mirror against 441 unpaired, so a budget of 800 falls between the two cliffs and the measured 7.4% against 99.0% at that budget is the predicted magnitude, not only the predicted sign.

task difficulty	mirror, recovery	unpaired, recovery	mirror FWER (null)	unpaired FWER (null)
homogeneous (400 invocations)	0.0 (≤ 0.7)	10.5 $\pm$ 1.1%	0.0 (≤ 0.7)	0.0 (≤ 0.7)
homogeneous (800 invocations)	7.4 $\pm$ 1.0%	99.0 $\pm$ 0.4%	0.0 (≤ 0.7)	0.0 (≤ 0.7)
Beta, conc. 4 (1600 invocations)	99.6 (≥ 98.4)	96.9 $\pm$ 0.6%	0.0 (≤ 0.7)	0.0 (≤ 0.7)
Beta, conc. 1 (1600 invocations)	98.6 $\pm$ 0.5%	73.6 $\pm$ 1.6%	0.0 (≤ 0.7)	1.0 $\pm$ 0.5%

What the procedure tests after it has decided something

Table 7. Proposition 4 read against the two cells built to probe it: 6400 invocations, 400 replications, heterogeneous tasks, $k = 5$ shown; the artifact carries all 18 intersection-null-true cells: the cond-null grid (one strong component plus one all-stage non-positive component, distinct from the main grid of Tables 4-5) over both component counts, three spreads and three budgets, with admission min/median/max 0.0/0.0/0.2% against the promised 5%. In the sign-change cell the tracked component helps before the first commitment and hurts after it, so it lies outside the intersection null and its admission rate is the price of the flip, not an error rate.

cell	tracked component vs the null	sequential screen admits	fixed-design marginal procedure admits
sign change (outside the null)	helps before commitment, hurts after	10.8%	97.2%
all-stage non-positive (null true)	helps in no reachable context	0.0%	0.0%

The marginal procedure's admissions in the sign-change cell are not errors under its own null; every one of them nonetheless ships a component that hurts the deployed configuration, which is the practical argument for the conditional estimand. The pinned admission superficially resembles the Resolution III alias rate of Table 3, and the resemblance is the paper's point: one is an announced change of estimand, the other is bias of a deterministic design under the null it claims to test. The corrected interim baseline is also run: the same eight looks as the naive arm, each tested at a Bonferroni spend over looks, holds the family-wise error (3.2% under the null at $k = 5$, heterogeneous, 6400) at a familiar price in power (58.5% recovery in the main-effect cell against 91.8% for the mirror e-process), and unlike the e-process it cannot be extended past its planned looks without a new spending function.

Both remedies a deployer might reach for are run in the same cell. Restarting the undecided processes at each commitment, with α spent $\alpha/2^e$ across epochs, lowers the flipped component's admission only from 10.8% to 8.8%; re-screening the undecided set afresh with the committed levels pinned gives the same 8.8%. The paths say why: essentially every admission of the flipping component happens at or before the first commitment, while its running effect is genuinely positive, so no post-commitment bookkeeping can prevent it. What a deployer can do is verify rather than bookkeep: a component admitted before a commitment it interacts with should be confirmed by a fresh fixed-sample comparison at the final committed configuration, which is exactly the post-decision check Section 7 runs live. The all-stage non-positive cell stays at 0.0% under the restart, as the theorem promises.

Judging a self-improvement loop

The screening problem does not only arise when a team hand-picks components. Self-improvement loops in the SkillOpt family, and prompt-evolution systems more broadly (Agrawal et al., 2025), generate bounded prompt edits and accept them on a held-out point comparison: run the champion and the variant, adopt if the variant scores higher. At the replay flip rate measured on our deployment that acceptance rule is close to a coin flip per candidate, and Table 9 measures what follows: with ten inert candidates the point rule adopts at least one in essentially every round, a per-candidate z-test without multiplicity control still fails in 21.8% (five candidates) to 37.5% (ten) of rounds, and the mirror screen at the same invocation budget stays at the nominal level while recovering the genuinely good edits. The guarantee of Proposition 5 is per proposal round; a loop that runs R rounds and wants one family-wise statement across all of them must spend α over rounds (α/R , or any positive summable schedule), and nothing in the construction obstructs that. Table 8 tabulates the division of labour with the measured evidence for each role. The proposal machinery and the judgment are separable: this procedure replaces only the acceptance rule.

Table 8. Division of labour in the composed loop, with the measured evidence for each role. The three existing systems keep their jobs; the judge is the missing stage.

stage	system	role kept	measured basis
propose	SkillOpt-family protocol (published representative: reflective prompt evolution, Agrawal et al., 2025)	bounded edits, protected regions, rejected-candidate memory	its acceptance rule fails by arithmetic: exact round-level false adoption 0.965 to 0.999 (Proposition 5)
execute	agent runtime (DeepSeek Harness in our live run)	tools, sessions, recovery; same-task paired replay	450 live mirrored pairs (900 invocations) completed with ledger resume, Section 7

context	NOOA-style preview handles	hands large objects to the proposer as handles	screened as a component rather than assumed; the digest switch in Section 7
judge	this procedure	admission and pruning with family-wise control at any stopping time	0.5 to 1.8% false adoption at equal budget (Table 9)

Table 9. Acceptance rules for proposed edits, 6400 invocations, 400 replications. False adoption = at least one edit with no effect adopted in the round; null = all candidates inert; live = two of the candidates carry a genuine effect of 0.15.

acceptance rule	false adoption, k=5 null	false adoption, k=10 null	false adoption, k=10 live	recovery, k=10 live
point comparison (adopt if better)	96.8 $\pm$ 0.9%	100.0 ($\geq$ 99.3)	99.2 $\pm$ 0.4%	95.4 $\pm$ 0.7%
point comparison, 2pp margin	75.8 $\pm$ 2.1%	98.2 $\pm$ 0.7%	95.8 $\pm$ 1.0%	88.4 $\pm$ 1.2%
per-candidate z-test, no correction	21.8 $\pm$ 2.1%	37.5 $\pm$ 2.4%	35.0 $\pm$ 2.4%	54.6 $\pm$ 1.8%
mirror screen (this paper)	1.8 $\pm$ 0.7%	0.8 $\pm$ 0.4%	0.5 ($\leq$ 1.6)	88.1 $\pm$ 1.1%

Where every method fails, and why that is the correct answer

In the negative-interaction scenario no method recovers the active components: the best arm reaches 7.4% at the largest budget. This is not a limitation of any procedure. The true mirror contrast in that scenario is 0.0047, roughly a sixth of the main-effect scenario, because the interaction cancels most of what the components contribute. Reporting the estimand alongside the power is what separates a method that fails from a setting in which there is nothing to find.

7. A live run on a production agent

The procedure is running, not proposed. The runtime is a current agent harness (DeepSeek Harness, headless), the model is fixed (claude-haiku-4-5), and the tasks are 358 labelled customer questions from 18 production shops, each with its own knowledge base and a deterministic, label-anchored grader: an answerable question must be answered with cited evidence, a question the knowledge base cannot answer must be declined or escalated. Five switches, all real harness components, are screened at once: a one-file index of the knowledge base (the preview-handle idea from context-management work), an instruction to search before answering, a mandatory self-check of claims against evidence, one worked example, and a guardrail that forbids answering without supporting evidence. Every row executes a configuration and its mirror on the same question, and every pair is appended to a ledger before it reaches the screen, so the run resumes after interruption; the figures below are the state of the ledger at build time, 450 mirrored pairs, of which 24.2% were discordant.

Table 10. Live verdicts at 450 pairs (admission threshold $k/\alpha = 100$). An anytime-valid screen may simply keep running; no error budget is spent by reporting this interim state.

switch	verdict	paired effect $\pm$ 95% anytime CS	evidence for	evidence against
digest	undecided (running)	+0.026 $\pm$ 0.184	3.77	1.00
search_hint	undecided (running)	+0.019 $\pm$ 0.184	1.49	0.80
reflect	undecided (running)	-0.026 $\pm$ 0.184	1.00	3.80
fewshot	undecided (running)	+0.001 $\pm$ 0.184	0.71	0.72
strict	pruned	-0.073 $\pm$ 0.273	1.00	115.91

One decision has been reached. The evidence-strictness guardrail was pruned as harmful at row 199, at which point its against-process stood at 115.9 against a threshold of 100. Its running paired effect at the decision was -0.073, with a stitched anytime confidence sequence of ± 0.273 at 95% (conservative, Hoeffding-type on bounded increments). That interval does not exclude zero: the universal-bound interval is far blunter than the betting process on the same data, and it is the e-process, which adapts its bets to the observed variance, that certified the pruning. A variance-adaptive confidence sequence would be tighter; we report the conservative one rather than none. The decision consumed 199 mirrored pairs,

398 model invocations, and 5.0 hours of wall time on the runtime; token counts were not logged and are reported as absent rather than estimated.

The pruning statement above is the harm process's own α -level statement, not an external confirmation, so an independent check was run after the decision: a fixed-sample paired comparison of the two configurations that differ only in the pruned guardrail (on against off, the other four switches held at the deployment's baseline, off) on 120 task draws, sampled without exclusion from the same 358-task pool the screen used (fresh outcomes, not a task split; a fixed seed disjoint from the screen's), task-paired, costing 2x that many invocations on top of the screen's 398, graded by the same deterministic grader. The guardrail-on arm passed 64 of 120 and the off arm 80 of 120; the mean paired difference was -0.133 with 95% interval [-0.210, -0.057]. The screen's paired effect is half the on-minus-off difference (Section 3), so its -0.073 corresponds to -0.146 on the confirmation's scale, inside the confirmation's interval. The confirmation is a different procedure from the one under test, which is its point; the two are reported side by side rather than merged.

The deployment also supplies the chooser of Section 6 with its first pilot on a response surface we did not write. Raw per-task means from few draws inflate the apparent between-task spread with binomial noise, and the criterion is monotone in that spread, so the estimate is corrected by variance decomposition: across the 92 tasks with at least two ledger pairs, the observed spread 0.141 decomposes into estimated binomial noise 0.031 and between-task variance 0.109, and shrinking the means accordingly moves the per-invocation log-growth ratio from 1.39 raw to 1.22 corrected (bootstrap over tasks, 95% interval 1.08-1.37, excluding one) - still recommending the mirror, the arm this run used. The ledger's rows also mix configuration levels rather than holding the champion, so this is a demonstration that the criterion is computable from a real pilot with its noise priced, not a validation of its choice, which would require running the unpaired arm live against it.

Two trends are worth naming because both run against the folklore that motivated the switches. The evidence is accumulating against the mandatory self-check and in favour of the knowledge-base index; the worked example sits near its futility floor with evidence below one in both directions, which is the honest reading of a switch that does nothing detectable at this budget. No verdict is claimed before a process crosses its threshold, and the guarantee is exactly that such interim readings can be taken at any time without inflating the error rate: that property, not any single verdict, is what the live run demonstrates.

8. What this does not establish

The method comparison rests on simulation whose parameters are measurements; the live run of Section 7 exercises the procedure itself, one deployment, one decision. The generative model is ours, and a reader is entitled to suspect a model that flatters its author. Three things are offered against that suspicion, none a substitute for the live run: the two parameters that drive difficulty are measured rather than chosen, the estimand of every scenario is published exactly rather than described, and the arms are separated so that each property is priced on its own. The reported grid contains dozens of cells and several near-boundary comparisons, and we have not corrected our own reporting for that multiplicity; the family sizes - seven criterion cells (one within noise), four perturbed settings, 24 null cells with one 1.4-standard-error excursion - let a reader price near-threshold agreements. Repeating the live run on a public verified terminal benchmark, where the tasks and graders are not ours, is the natural extension. Two further checks bound the reading. Under an exact null the plug-in bet shrinks toward zero, so the bound is far from binding: a boundary cell (homogeneous, zero effect, 25,600 invocations, 1,000 replications) admits in 0.4% of runs against the promised 5% - the safety margins are a property of this bet family, not sharpness. And both calibrated parameters were perturbed (replay flip 0.06-0.18, failure share 0.40-0.60): the Resolution III alias admission stays at 91-97%, the e-process keeps inert admissions at or below 1.0%, and its recovery moves over 71.2-98.5% - the conclusions survive, the power does not always.

Table 11. Objections we consider strongest, and where each is addressed.

objection	status	response
The mirror estimator is not new	granted	Antithetic variates, simultaneous perturbation and mirrored sampling all use it. The contribution is the per-component sequential admission built on it, and the measurement of how the standard alternatives fail.
Synthetic evidence for the method comparison	partly granted	The comparison of procedures is simulation with measured parameters and published estimands; the procedure itself also ran live (Section 7) and produced one error-controlled pruning on a production agent.

Mirroring can produce a configuration nobody would deploy	partly granted	Components may be pinned; only the flipped subset needs to be mirrored, at the cost of screening a subset at a time.
The estimand averages over the other components	granted	Stated where it is derived. Pin the rest to obtain the deployed-configuration effect.
Two invocations per row is wasteful	granted on binary verifiers	The unpaired one-invocation e-process with predictable centring is valid and is the default recommendation; the mirror is chosen only when the per-invocation log-growth ratio of Section 6 exceeds one, which on binary verifiers requires strong task-difficulty spread and on graded verifiers is typical.
The bet size has free parameters	answered	Nine settings of the cap and the bet stabiliser c (distinct from the futility floor $f = 0.05$ of Algorithm 1) were run at both component counts. Under the global null the family-wise error stayed at or below 1.5% in every one of the eighteen cells, mean 0.4%. Recovery moved with the stabiliser, from 75.0% to 93.2% at ten components, so the knob buys power and cannot buy validity. The headline cells use the pre-registered default $c = 0.1$, kept for guard margin; the best-recovery ablation setting ($c = 0.05$) is reported here, not silently adopted.
A union bound over components is conservative	granted	An e-BH variant controlling the false discovery rate is the obvious extension.
Admit or prune is not a shipping decision without an effect size	partly granted	The screen reports the running mean paired contrast per component alongside the verdict; a full anytime-valid confidence sequence per component is future work.
The motivating citations are recent preprints	granted	The harness-effect literature is months old; the build resolves identifiers but cannot confer peer review on them. The claims we rely on quantitatively are our own measurements.

9. Conclusion

Teams now tune agent harnesses with many switches, and the affordable ways to decide what stays fail measurably: a cheap fractional design names an inert component nine times out of ten under interaction, and unadjusted interim looks inflate the error rate by an order of magnitude. Mirror screening removes both failures with no design matrix, no enumeration and no fixed sample size, at a measured, decomposed price. The estimator is old; the guarantee, application and measurement of the alternatives are what we contribute.

Table 12. What a practitioner takes from this paper, as rules.

situation	rule	measured basis
choosing which harness switches stay	randomise levels per task, one e-process per switch, stop whenever you like	FWER at nominal in every null cell (admission and pruning each at their own α ; acting on both reads 2α); interim looks without this inflate error to 29-42% at 6400 invocations
tempted by a fractional design	do not: randomisation gives the same cheapness without the alias	88-94% wrong-component admission under interaction
accepting proposed prompt edits	never on a point comparison, whatever the margin	0.96-0.999 round-level false adoption, computed exactly
pair with mirror or run unpaired	estimate between-task spread from a pilot; pair only when the log-growth ratio exceeds one, always on graded verifiers	six of seven paired-versus-unpaired cells resolved as predicted, one within noise (Tables 5-6)
reading verdicts after early commitments	the guarantee is the intersection null; a component that helps in no committed context is admitted at most α ; a component whose effect may flip after a commitment carries no guarantee and should be re-screened with the others pinned at the committed levels	worst 0.2% across 18 cells with the null true

Table 13. Family-wise error under the global null, heterogeneous tasks, by arm and budget: the cells behind Figures 1 and 3 and the interim-look numbers quoted in the abstract and Section 6. The e-process rows come from the mirror grid, whose null world is the same by construction; the small differences from the design-grid arms are sampling error at 400 replications. One cell sits above nominal - fold-over at $k = 10$, 6400 reads 6.8 ± 1.3 against the nominal 5 - which is 1.4 standard errors and, with 24 cells tabulated, within expectation.

arm	budget	FWER, $k=5$	FWER, $k=10$
fold-over, unadjusted interim looks	400	12.0 ± 1.6	23.2 ± 2.1
fold-over, unadjusted interim looks	1600	22.8 ± 2.1	28.2 ± 2.3
fold-over, unadjusted interim looks	6400	29.0 ± 2.3	41.8 ± 2.5
champion paired, e-process	400	0.8 ± 0.4	0.8 ± 0.4
champion paired, e-process	1600	1.5 ± 0.6	1.0 ± 0.5
champion paired, e-process	6400	2.8 ± 0.8	2.0 ± 0.7
full factorial, fixed sample	400	2.2 ± 0.7	3.5 ± 0.9
full factorial, fixed sample	1600	3.2 ± 0.9	2.2 ± 0.7
full factorial, fixed sample	6400	4.8 ± 1.1	2.0 ± 0.7
fold-over (Resolution IV), fixed sample	400	2.8 ± 0.8	3.8 ± 0.9
fold-over (Resolution IV), fixed sample	1600	3.2 ± 0.9	4.0 ± 1.0
fold-over (Resolution IV), fixed sample	6400	3.8 ± 0.9	6.8 ± 1.3
mirror pairing, e-process (this paper)	400	$0.0 (\leq 0.7)$	$0.0 (\leq 0.7)$
mirror pairing, e-process (this paper)	1600	$0.2 (\leq 1.2)$	$0.5 (\leq 1.6)$
mirror pairing, e-process (this paper)	6400	$0.0 (\leq 0.7)$	$0.5 (\leq 1.6)$

10. Impact statement

This work aims to make component decisions in deployed LLM agent systems error-controlled: the direct consequence is fewer superstitious components shipped, fewer harmful ones retained, and less compute spent on screens that must be rerun because their first run carried no guarantee. The main risk is misuse of the guarantee's name: the family-wise bound covers the admission decision, not the validity of the outcome measure it is fed, and a screen run on a broken verifier will certify nonsense with confidence. Section 8 states this boundary. The live run uses labelled customer-service questions processed under the deployment's existing data terms; no personal data leaves the deployment.

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23. **DeepSeek Harness (software).** Agent runtime used in the live run. npm package @deepseek-ai/dsh, version 0.1.0-rc.7, headless profile, model held fixed.
24. **NOOA (software).** NVIDIA object-oriented agents library, version 0.0.8; the knowledge-base index switch follows its preview-handle context pattern.
25. **Skill-optimisation loops (software family).** Bounded prompt-edit loops with held-out point-comparison acceptance, as implemented in current open-source skill optimisers; a published representative of the proposal side is reflective prompt evolution (Agrawal et al., 2025, arXiv:2507.19457). The acceptance rule is the one refuted exactly in Section 6.
26. **Hammersley and Morton, 1956.** A new Monte Carlo technique: antithetic variates. Mathematical Proceedings of the Cambridge Philosophical Society 52(3).
27. **Spall, 1992.** Multivariate stochastic approximation using a simultaneous perturbation gradient approximation. IEEE Transactions on Automatic Control 37(3).
28. **Box, Hunter and Hunter, 2005.** Statistics for Experimenters, 2nd edition. Wiley. Fold-over of a Resolution III design.
29. **Satterthwaite, 1959.** Random balance experimentation. Technometrics 1(2). The prior art for reading main effects off randomly drawn factor levels, together with the published critiques in the same issue.
30. **Robbins, 1970.** Statistical methods related to the law of the iterated logarithm. Annals of Mathematical Statistics 41(5). Power-one tests.
31. **Ville, 1939.** Etude critique de la notion de collectif. Gauthier-Villars.

Artefacts. Every number in this manuscript is read at build time from the experiment outputs: *exp_screen.json* and *exp_mirror.json* for the simulations, *exp_v2.json* for the calibration, restart, boundary and sensitivity cells, *e2_confirm.json* for the held-out live confirmation, *targets.json* for the estimands, *theory_check.py* for the deterministic verification of Propositions 1 and 2, and *citations.json* for the arXiv resolution of every reference. Rebuilding this document with a changed run changes the text. The unpaired fixed-sample arm was regenerated after correcting its statistic from the uncentred $\bar{x}_i y$ to the centred $\bar{x}_i(y - \bar{y})$ its sequential twin uses; the uncentred version had understated that arm by 16 points. An ICML two-column LaTeX build with an 8-page main body (*tex/paper.tex*, compiled with the official icml2025 style) accompanies the artefact; in that build the floats named next sit in the appendix, with numbering unchanged; the appendix holds Tables 1, 2 and 5-14 and Figures 2-8, and the main body keeps Figure 1, Algorithm 1 and Tables 3-4.